Determine whether robust Stackelberg equilibrium admits a polynomial-time approximation scheme

ChatGPT (`gpt-5.6-sol`)*

Solution generated on 2026-08-24

Abstract

We give a complete solution for every positive robustness parameter δ when both players' payoffs are normalized to $[0,1]$. For every fixed, efficiently representable $0 < \delta \leq 1$, we prove that, under the deterministic exponential-time hypothesis for PPAD, computing a leader strategy within a fixed additive constant of the optimal robust Stackelberg objective requires

$$(m+n)^{(\log(m+n))^{1-o(1)}}$$

time, where m and n are the numbers of leader and follower actions. Consequently, no polynomial-time approximation scheme exists in this range. The reduction is from constant-approximate bimatrix Nash equilibrium and produces instances whose exact robust optimum is 1/2; it therefore proves hardness of producing a leader strategy, not merely hardness of estimating the optimum value. The endpoint $\delta = 1$ is included, and its inclusion depends essentially on the strict inequality in the definition of a δ-best response. For $\delta > 1$, every follower action is always a strict δ-best response, and the exact leader optimum is computable in polynomial time by linear programming. Thus no gap remains for $\delta > 0$.

1 The open problem

This paper addresses an open problem posed by Jiarui Gan, Minbiao Han, Jibang Wu, Haifeng Xu in *Robust Stackelberg Equilibria* [1] (Economics and Computation (EC), 2023), Section 1.1 (Our contributions), page 4, footnote 2 and surrounding paragraph; and Section 4.2 (A QPTAS for δ-RSE), page 21, paragraph after introducing the QPTAS.. The website catalogue entry is problem 10 (`W4383533535_p0`) of the Open Problems in Operations Research collection.

> *Decide which of the following holds for computing an additive approximation to δ-robust Stackelberg equilibrium in general finite Stackelberg games:*
>
> *1) There exists a polynomial-time approximation scheme (PTAS): for every $\varepsilon > 0$, there is an algorithm running in time* $\mathrm{poly}(m,n) \cdot f(1/\varepsilon)$ *(for some function f) that outputs an ε-optimal δ-RSE leader strategy $\hat{x}$, i.e.,* $\min_{j \in \mathrm{BR}_\delta(\hat{x})} u_l(\hat{x}, j) \geq u_{\mathrm{RSE}}(\delta) - \varepsilon$.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260824_033055_c4359ada, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

2) No such PTAS exists (under a standard complexity assumption), i.e., the QPTAS is essentially optimal up to quasi-polynomial factors, for example by proving hardness of achieving a constant additive approximation (or ruling out PTAS) for δ-RSE under an assumption such as the Exponential Time Hypothesis.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

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2 δ-Best Responses

3 Problem and main result

Consider a finite two-player Stackelberg game. The leader has pure-action set $[m] = \{1, \dots, m\}$, the follower has pure-action set $[n] = \{1, \dots, n\}$, and both payoff matrices have rational entries in $[0, 1]$. For a leader strategy $z \in \Delta_m$ and a follower action $a \in [n]$, write

$$u_r(z, a) = \sum_{s=1}^{m} z_s \, u_r(s, a), \qquad r \in \{l, f\}.$$

For $\delta > 0$, the follower's strict δ-best-response set is

$$\mathrm{BR}_\delta(z) = \left\{ a \in [n] : u_f(z, a) > \max_{h \in [n]} u_f(z, h) - \delta \right\}.$$

The robust leader objective is

$$F_\delta(z) = \min_{a \in \mathrm{BR}_\delta(z)} u_l(z, a), \qquad \mathrm{OPT}_\delta = \max_{z \in \Delta_m} F_\delta(z).$$

An additive η-optimal leader strategy is a strategy z satisfying

$$F_\delta(z) \geq \mathrm{OPT}_\delta - \eta.$$

We use the deterministic PPAD–ETH in the form asserting that EndOfALine instances of size s require $2^{\widetilde{\Omega}(s)}$ deterministic time.

Theorem 1 (Complete classification)**.** *The following statements hold for normalized rational-payoff games.*

1. *For every fixed, efficiently representable δ with $0 < \delta \leq 1$, there is an absolute constant $\eta_{\mathrm{R}} > 0$, independent of the game size and of δ, such that, under the deterministic PPAD–ETH, computing an additive η_{R}-optimal leader strategy requires*

$$(m+n)^{(\log(m+n))^{1-o(1)}}$$

   time in the worst case. In particular, there is no deterministic polynomial-time approximation scheme.

2. *For every $\delta > 1$, the exact value OPT_δ and an optimal leader strategy can be computed in polynomial time by rational linear programming.*

The theorem is a complete solution for the stated domain $\delta > 0$. The first conclusion concerns deterministic algorithms; the analogous conclusion for randomized algorithms follows under the corresponding randomized version of the PPAD–ETH.

4 The Nash-equilibrium hardness theorem

For a bimatrix game $(R, C) \in [0,1]^{d\times d}$, a pair $(p,q) \in \Delta_d \times \Delta_d$ is an ε-approximate Nash equilibrium in the expected-regret sense if

$$\max_{k\in[d]} R_k q - p^\top R q \le \varepsilon \qquad \text{and} \qquad \max_{\ell\in[d]} p^\top C_{\cdot\ell} - p^\top C q \le \varepsilon,$$

where R_k denotes row k of R and $C_{\cdot\ell}$ denotes column ℓ of C.

A pair (p,q) is an ε-well-supported Nash equilibrium if

$$R_i q \ge R_k q - \varepsilon \quad \text{for every } i \in \operatorname{supp}(p) \text{ and every } k \in [d],$$

and

$$p^\top C_{\cdot j} \ge p^\top C_{\cdot\ell} - \varepsilon \quad \text{for every } j \in \operatorname{supp}(q) \text{ and every } \ell \in [d].$$

Every ε-well-supported Nash equilibrium is an ε-approximate Nash equilibrium. Indeed, averaging the supported-row inequalities with weights p_i gives

$$p^\top R q \ge \max_k R_k q - \varepsilon,$$

and averaging the supported-column inequalities with weights q_j gives

$$p^\top C q \ge \max_\ell p^\top C_{\cdot\ell} - \varepsilon.$$

The hardness result on which our reduction rests is the following normalized version of Rubinstein's theorem.

Theorem 2 (Rubinstein)**.** *Assume the deterministic PPAD–ETH. There exists an absolute constant $\varepsilon_{\mathrm{N}} \in (0,1)$ such that computing an ε_{N}-approximate Nash equilibrium, in the expected-regret sense, of a rational $d \times d$ bimatrix game with payoffs in $[0,1]$ requires*

$$d^{(\log d)^{1-o(1)}}$$

deterministic time.

This is Theorem 1.2 of Rubinstein [4]. The hard instances have two players, the same number d of actions for each player, constant-bounded payoffs, and rational coefficients of polynomial bit complexity. If the theorem is stated with payoffs in another fixed bounded interval, applying a fixed positive affine transformation to each player's payoffs maps that interval to $[0,1]$, preserves best responses, and scales all regrets by a fixed positive constant; this only changes the existential constant ε_{N}. Thus all hypotheses of the normalized form in Theorem 2 are satisfied by the source instances used below.

5 Reduction

Fix an efficiently representable δ with $0 < \delta \le 1$, and let $(R, C) \in [0,1]^{d\times d}$ be a rational bimatrix game. The construction below is valid for every $d \ge 1$.

The leader has the $2d$ pure actions

$$L_1, \ldots, L_d, \qquad Q_1, \ldots, Q_d.$$

Write a leader strategy as $z = (x, y)$, where

$$x, y \in \mathbb{R}^d_{\geq 0}, \qquad \alpha = \sum_{i=1}^{d} x_i, \qquad \beta = \sum_{j=1}^{d} y_j = 1 - \alpha.$$

The follower has two balance actions b_L, b_Q, a row-comparison action ρ_{ik} for every ordered pair $i, k \in [d]$, and a column-comparison action $\kappa_{j\ell}$ for every ordered pair $j, \ell \in [d]$.

For every leader action s, define

$$u_f(s, b_L) = u_f(s, b_Q) = \delta.$$

For the row-comparison actions, define

$$u_f(s, \rho_{ik}) = \begin{cases} \delta, & s = L_i, \\ 0, & \text{otherwise}, \end{cases}$$

and for the column-comparison actions,

$$u_f(s, \kappa_{j\ell}) = \begin{cases} \delta, & s = Q_j, \\ 0, & \text{otherwise}. \end{cases}$$

Because $0 < \delta \leq 1$, all follower payoffs lie in $[0, 1]$.

The leader payoffs against the balance actions are

$$u_l(L_s, b_L) = 1, \qquad u_l(Q_t, b_L) = 0,$$

and

$$u_l(L_s, b_Q) = 0, \qquad u_l(Q_t, b_Q) = 1.$$

For every $i, k, s, t \in [d]$, define the leader payoffs against ρ_{ik} by

$$u_l(L_s, \rho_{ik}) = \frac{1}{2}, \qquad u_l(Q_t, \rho_{ik}) = \frac{1}{2} + \frac{R_{it} - R_{kt}}{2}.$$

For every $j, \ell, s, t \in [d]$, define the leader payoffs against $\kappa_{j\ell}$ by

$$u_l(Q_t, \kappa_{j\ell}) = \frac{1}{2}, \qquad u_l(L_s, \kappa_{j\ell}) = \frac{1}{2} + \frac{C_{sj} - C_{s\ell}}{2}.$$

Since every entry of R and C lies in $[0, 1]$, every difference appearing above lies in $[-1, 1]$. Consequently, all constructed leader payoffs lie in $[0, 1]$.

Lemma 1 (Exact activation identity)**.** *For every leader strategy $z = (x, y)$,*

$$\mathrm{BR}_\delta(z) = \{b_L, b_Q\} \cup \{\rho_{ik} : x_i > 0\} \cup \{\kappa_{j\ell} : y_j > 0\}.$$

Proof. The expected follower payoffs are

$$u_f(z, b_L) = u_f(z, b_Q) = \delta, \qquad u_f(z, \rho_{ik}) = \delta x_i, \qquad u_f(z, \kappa_{j\ell}) = \delta y_j.$$

The balance actions attain payoff δ, and every comparison action has payoff at most δ. Hence

$$\max_a u_f(z, a) = \delta,$$

and the strict δ-best-response condition becomes

$$u_f(z, a) > \delta - \delta = 0.$$

Both balance actions satisfy this inequality because $\delta > 0$. Moreover,

$$\delta x_i > 0 \iff x_i > 0, \qquad \delta y_j > 0 \iff y_j > 0,$$

which proves the identity.

At the endpoint $\delta = 1$, an unused comparison action has payoff exactly zero and the threshold is also zero. Such an action is excluded because the definition uses the strict inequality $u_f(z, a) > 0$, so the same identity remains valid. $\square$

The expected leader payoffs against the balance actions are

$$u_l(z, b_L) = \alpha, \qquad u_l(z, b_Q) = \beta. \tag{1}$$

For a row-comparison action,

$$\begin{aligned} u_l(z, \rho_{ik}) &= \sum_{s=1}^{d} x_s \frac{1}{2} + \sum_{t=1}^{d} y_t \left(\frac{1}{2} + \frac{R_{it} - R_{kt}}{2} \right) \\ &= \frac{1}{2} + \frac{1}{2}(R_i - R_k) y. \end{aligned} \tag{2}$$

For a column-comparison action,

$$\begin{aligned} u_l(z, \kappa_{j\ell}) &= \sum_{t=1}^{d} y_t \frac{1}{2} + \sum_{s=1}^{d} x_s \left(\frac{1}{2} + \frac{C_{sj} - C_{s\ell}}{2} \right) \\ &= \frac{1}{2} + \frac{1}{2} x^\top (C_{\cdot j} - C_{\cdot \ell}). \end{aligned} \tag{3}$$

6 The exact robust optimum

We use the following standard existence theorem.

Theorem 3 (Nash). *Every finite normal-form game has a mixed-strategy Nash equilibrium.*

The theorem applies to (R, C) because each player has the finite action set $[d]$, each mixed-strategy space is a nonempty compact convex simplex, and the expected payoffs are continuous bilinear functions of the mixed strategies. These are precisely the finite-game hypotheses of Nash's existence theorem [3].

Proposition 1. *Every Stackelberg instance constructed above satisfies*

$$\mathrm{OPT}_\delta = \frac{1}{2}.$$

Proof. By Lemma 1, both balance actions are active for every leader strategy. Equation (1) therefore gives

$$F_\delta(z) \le \min\{\alpha, \beta\} \le \frac{\alpha + \beta}{2} = \frac{1}{2},$$

so $\mathrm{OPT}_\delta \le 1/2$.

For the reverse inequality, Theorem 3 provides an exact mixed Nash equilibrium (p, q) of (R, C). Define

$$z = \left(\frac{p}{2}, \frac{q}{2}\right).$$

Then $\alpha = \beta = 1/2$, so each balance action gives the leader payoff 1/2.

If $p_i > 0$, the support property of an exact Nash equilibrium gives

$$R_i q \ge R_k q \qquad \text{for every } k \in [d].$$

Indeed, if a supported pure action had payoff strictly below another pure action, moving its positive probability to the better action would strictly increase the row player's payoff. Since $x_i = p_i/2 > 0$, Lemma 1 shows that every ρ_{ik} is active, and Equation (2) yields

$$u_l(z, \rho_{ik}) = \frac{1}{2} + \frac{1}{4}(R_i q - R_k q) \ge \frac{1}{2}.$$

Similarly, if $q_j > 0$, the column player's Nash support condition is

$$p^\top C_{\cdot j} \ge p^\top C_{\cdot \ell} \qquad \text{for every } \ell \in [d].$$

Since $y_j = q_j/2 > 0$, every $\kappa_{j\ell}$ is active, and Equation (3) gives

$$u_l(z, \kappa_{j\ell}) = \frac{1}{2} + \frac{1}{4} p^\top (C_{\cdot j} - C_{\cdot \ell}) \ge \frac{1}{2}.$$

Thus every active follower action gives the leader payoff at least 1/2, while both balance actions give exactly 1/2. Hence

$$F_\delta(z) = \frac{1}{2},$$

which proves $\mathrm{OPT}_\delta \ge 1/2$. Combining the two bounds proves the claim. □

Nash's theorem is used here only to establish the value attained by the robust instance. The reduction does not use an equilibrium as part of the computational construction and is therefore not circular.

7 Decoding a near-optimal leader strategy

Proposition 2 (Approximation decoding)**.** *Let $0 \le \eta < 1/2$, and suppose that a leader strategy $z = (x, y)$ in the constructed game satisfies*

$$F_\delta(z) \ge \frac{1}{2} - \eta.$$

Then a $\dfrac{4\eta}{1 - 2\eta}$-well-supported Nash equilibrium of the source game (R, C) can be obtained in polynomial time.

Proof. Because both balance actions are active, Equation (1) and the assumed lower bound on $F_\delta(z)$ imply

$$\alpha \geq \frac{1}{2} - \eta, \qquad \beta \geq \frac{1}{2} - \eta.$$

Since $\eta < 1/2$, both α and β are strictly positive. Consequently,

$$p = \frac{x}{\alpha}, \qquad q = \frac{y}{\beta}$$

are well-defined mixed strategies.

Fix any $i \in \mathrm{supp}(p)$. Then $x_i > 0$, so Lemma 1 implies that ρ_{ik} is active for every $k \in [d]$. Since $F_\delta(z)$ is the minimum leader payoff over all active actions,

$$u_l(z, \rho_{ik}) \geq \frac{1}{2} - \eta.$$

Using $y = \beta q$ in Equation (2) gives

$$\frac{1}{2} + \frac{\beta}{2}(R_i q - R_k q) \geq \frac{1}{2} - \eta,$$

and therefore

$$R_i q \geq R_k q - \frac{2\eta}{\beta}.$$

The lower bound $\beta \geq 1/2 - \eta$ implies

$$\frac{2\eta}{\beta} \leq \frac{2\eta}{1/2 - \eta} = \frac{4\eta}{1 - 2\eta}.$$

Thus, for every $i \in \mathrm{supp}(p)$ and every $k \in [d]$,

$$R_i q \geq R_k q - \frac{4\eta}{1 - 2\eta}. \tag{4}$$

Now fix any $j \in \mathrm{supp}(q)$. Then $y_j > 0$, so every $\kappa_{j\ell}$ is active. Equation (3), with $x = \alpha p$, gives

$$\frac{1}{2} + \frac{\alpha}{2} p^\top (C_{\cdot j} - C_{\cdot \ell}) \geq \frac{1}{2} - \eta,$$

and hence

$$p^\top C_{\cdot j} \geq p^\top C_{\cdot \ell} - \frac{2\eta}{\alpha}.$$

Using $\alpha \geq 1/2 - \eta$, we obtain, for every $j \in \mathrm{supp}(q)$ and every $\ell \in [d]$,

$$p^\top C_{\cdot j} \geq p^\top C_{\cdot \ell} - \frac{4\eta}{1 - 2\eta}. \tag{5}$$

Inequalities (4) and (5) are exactly the well-supported Nash conditions with error $4\eta/(1-2\eta)$. $\square$

Let ε_{N} be the constant from Theorem 2, and define

$$\eta_{\mathrm{R}} = \frac{\varepsilon_{\mathrm{N}}}{4 + 2\varepsilon_{\mathrm{N}}}. \tag{6}$$

This is a fixed positive constant smaller than $1/2$. Moreover,

$$1 - 2\eta_{\mathrm{R}} = \frac{4}{4 + 2\varepsilon_{\mathrm{N}}}, \qquad \text{and hence} \qquad \frac{4\eta_{\mathrm{R}}}{1 - 2\eta_{\mathrm{R}}} = \varepsilon_{\mathrm{N}}.$$

By Proposition 1, an additive η_{R}-optimal leader strategy satisfies

$$F_\delta(z) \geq \mathrm{OPT}_\delta - \eta_{\mathrm{R}} = \frac{1}{2} - \eta_{\mathrm{R}}.$$

Proposition 2 therefore converts it into an ε_{N}-well-supported Nash equilibrium, and hence into an ε_{N}-approximate Nash equilibrium in the expected-regret sense required by Theorem 2.

8 Size, encoding, and hardness transfer

The constructed Stackelberg game has $m = 2d$ leader actions and $n = 2 + 2d^2$ follower actions. Its two payoff matrices contain

$$mn = 2d(2 + 2d^2) = 4d + 4d^3 = O(d^3)$$

entries. The construction consists only of copying entries, subtracting two entries, dividing by two, and inserting the fixed numbers 0, 1/2, 1, and δ. Since the source instance has rational entries of polynomial bit complexity and δ is fixed and efficiently representable, every constructed payoff has polynomial bit complexity. The construction and the normalization

$$p = x/\alpha, \qquad q = y/\beta$$

can therefore be performed in polynomial time.

Let

$$D = m + n = 2d^2 + 2d + 2 = \Theta(d^2).$$

Since $\log D = \Theta(\log d)$, constants introduced by replacing d with $D^{1/2}$ are absorbed by the $o(1)$ term, and

$$d^{(\log d)^{1-o(1)}} = D^{(\log D)^{1-o(1)}}$$

in the usual asymptotic sense of this lower-bound notation.

Suppose there were an algorithm that, on every constructed game, returned an additive η_{R}-optimal leader strategy in time asymptotically below

$$D^{(\log D)^{1-o(1)}}.$$

Constructing the game, applying that algorithm, and performing the decoding of Proposition 2 would compute an ε_{N}-approximate Nash equilibrium of the source game faster than

$$d^{(\log d)^{1-o(1)}}.$$

The $O(d^3)$ construction cost and the polynomial-time decoding cost do not affect this quasi-polynomial lower bound. This contradicts Theorem 2, and the first part of Theorem 1 follows.

In particular, a polynomial-time approximation scheme could be run at the fixed accuracy η_{R}, giving a polynomial-time algorithm and contradicting the same lower bound. A fortiori, no algorithm with running time

$$\mathrm{poly}(m, n)\, f(1/\eta)$$

can provide the required additive guarantee for every $\eta > 0$.

This is genuinely strategy-output hardness. Proposition 1 shows that every instance produced by the reduction has the publicly known value

$$\mathrm{OPT}_\delta = \frac{1}{2}.$$

Thus the reduction cannot be interpreted merely as hardness of estimating the optimal objective value.

9 Exact solvability for $\delta > 1$

Assume now that $\delta > 1$, and consider an arbitrary normalized Stackelberg game. For every leader strategy z and follower action a,

$$u_f(z, a) \geq 0.$$

Also,

$$\max_h u_f(z, h) \leq 1, \qquad \text{so} \qquad \max_h u_f(z, h) - \delta < 0.$$

It follows that

$$u_f(z, a) > \max_h u_f(z, h) - \delta$$

for every follower action a. Therefore

$$\mathrm{BR}_\delta(z) = [n] \qquad \text{for every } z \in \Delta_m,$$

and the robust objective becomes

$$F_\delta(z) = \min_{j \in [n]} \sum_{i=1}^{m} z_i \, u_l(i, j).$$

Consequently, its exact optimum is represented by the linear program

$$\begin{aligned} \text{maximize} \quad & t \\ \text{subject to} \quad & \sum_{i=1}^{m} z_i \, u_l(i, j) \geq t, \quad && j = 1, \ldots, n, \\ & \sum_{i=1}^{m} z_i = 1, \\ & z_i \geq 0, && i = 1, \ldots, m. \end{aligned}$$

This program is feasible because Δ_m is nonempty, and its objective is bounded between 0 and 1.

The polynomial-time theorem for rational linear programming states that a rational linear program can be optimized, and an exact rational optimal solution returned, in time polynomial in the binary encoding length of its variables, constraints, and coefficients [2]. Its hypotheses hold here: the program has $m+1$ variables, $n+1$ substantive constraints together with nonnegativity constraints, rational coefficients inherited from the input payoff matrix, a nonempty feasible region, and a bounded objective. Hence both OPT_δ and an exact optimal leader strategy are computable in polynomial time.

10 Conclusion

For normalized payoffs and every positive δ, the classification is

$0 < \delta \le 1:$	Under the deterministic PPAD–ETH, a fixed constant-additive strategy approximation requires $(m+n)^{(\log(m+n))^{1-o(1)}}$ time. Hence no deterministic PTAS exists.
$\delta > 1:$	The exact robust optimum and an optimal leader strategy are computable in polynomial time by linear programming.

The strict response convention is essential at $\delta = 1$: actions whose follower payoff is exactly zero are excluded when the response threshold is zero. Thus the hard range includes the endpoint $\delta = 1$, while the transition to exact polynomial-time solvability occurs strictly above 1. This establishes a complete solution, rather than partial progress, for the stated domain $\delta > 0$.

References

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